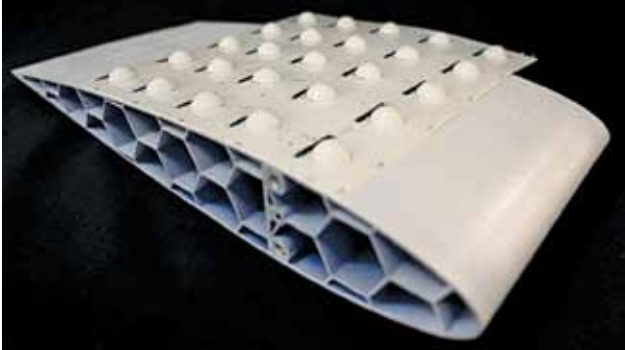




Retractable dome could save drones from data drowning dangers

The efficiency of drones is affected by a problem called 'data drowning' as too much data from their sensors affects their full flight capability. This prevents them from flying safely in certain situations. To overcome this issue, researchers at Purdue University and the University of Tennessee, Knoxville



Patterns of invertible domes on a drone's wings would give it an approach to bear in mind in microseconds what harmful situations really feel like and react rapidly (Source: <https://news8plus.com>)

have developed a small retractable dome inspired by a soda cup's to-go lid, arranged in the form of an array to provide memory cues to a drone wing for dangerous conditions. The dome pops up and down at specific parts of the wings to indicate to the drone's control system that the drone wing is encountering a threatening pressure pattern. As this dome consists of only two states—popped up or popped down—these states can act like zeros and ones to create spatial patterns for building associative memory.

Slip ring assembly for next-gen high-power thruster satellites

Researchers from the Swiss Plasma Center (SPC) of EPFL and Beyond Gravity have implemented a slip ring assembly



Scientists at the Swiss Plasma Center at EPFL, in collaboration with Beyond Gravity and with the support of the ESA, have developed a slip ring assembly that can more than triple the operational voltage of new-generation, high-voltage satellites (Source: <https://actu.epfl.ch>)

bly that increases voltages in satellites, elevating the range from the current 28-100 volts to 300-600 volts, resulting in increased robustness against electrical failures. The slip ring transfers electrical power to electrical power systems—such as onboard computers, traveling wave tubes, data-gathering equipment, and power thrusters—by using sliding electrical contacts comprising a rotating ring and a stationary brush—the wiper. As the system rotates in a satellite, the wiper slides along the ring's surface, transferring electrical current at low voltage. This slip ring design could be implemented in the next generation of high-power thruster satellites.

Button-size antenna provides data transfer 100x faster than Bluetooth

Srikanta Pal, a professor of electronics and communication engineering at Birla Institute of Technology (BIT) Mesra, and Mrinmoy Chakraborty, a Ph.D. scholar in the same department at BIT Mesra, have developed a button-size wireless antenna that is 100 times faster than Bluetooth and can be used for office automation, including in schools, hospitals, and other entities. It would also make internal connectivity safe by enabling high-speed communication that is not dependent on the internet and eliminating wires and cables for short-range (less than 100m) wireless personal area networks (WPANs). This small antenna can transfer data or files up to a few gigabytes within seconds.

Smart glass encodes data into the sunlight passing through it

KAUST researchers have designed a smart glass system that encodes data into the sunlight passing through it, which is detected and decoded by devices in the room. The system is configured with a light modulator embedded in the glass surface and an in-room receiver. The

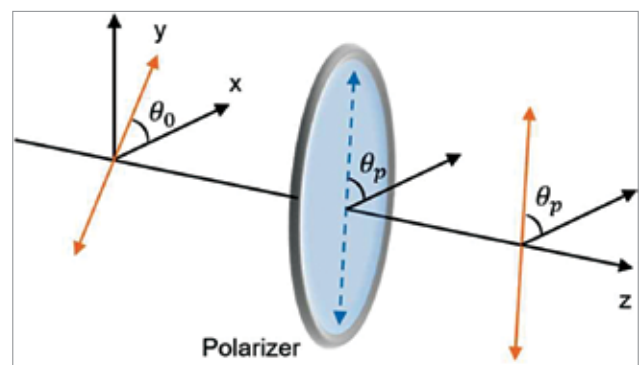


Illustration of the polarizer effect on the polarized light (Source: <https://techxplare.com>)